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MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

First Semester, September - October, 2025

Paper Code: ES-119**Subject: Manufacturing Processes****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1, which is compulsory and any two more questions from remaining.**

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Discuss Ductility and Toughness with appropriate physical definitions.	2	CO1
(b)	What is the purpose of using flux in welding processes?	2	CO2
(c)	Why is the riser used in the gating system? Explain its importance in solidifying metal.	2	CO1
(d)	What is edge preparation in welding, and under what conditions does it become necessary?	2	CO2
(e)	Discuss any two pattern allowances considered in sand casting.	2	CO1
2 (a)	Explain the working principle of the Hot Chamber Die Casting Process with a neat sketch. Give its key industrial applications.	6	CO1
(b)	Discuss the various types of electrodes used in Electric Arc Welding and their functions.	4	CO2
3 (a)	Differentiate between Direct Current Straight Polarity (DCSP) and Direct Current Reverse Polarity (DCRP) in Electric Arc Welding.	6	CO2
(b)	Describe any four desirable properties of moulding sand used in foundry casting.	4	CO1
4 (a)	Describe, along with neat sketches, the different types of flames in Gas Welding and state the practical application of each.	6	CO2
(b)	Explain any four casting defects commonly encountered. Also, state their causes and remedies.	4	CO1

****** End of Question Paper ******